The retrieval pipeline begins by indexing chunked representations into a vector store. Each chunk maintains explicit provenance including document identity and page numbers.

When queries arrive at the system, hybrid retrieval combines dense vector embeddings with sparse keyword matching. Reciprocal rank fusion merges the ranked candidates into an initial candidate set.

Cross-encoder reranking provides fine-grained relevance scoring across the candidate pool. The top-ranked passages are presented to the generation engine with verifiable citations.